

VENKATA RAMA KRISHNA DULAM

AI SYSTEMS & AGENT DEVELOPER | DOCUMENT INTELLIGENCE & RAG | SECURITY

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PROFILE

AI-focused Computer Science and Engineering student with hands-on experience building agentic systems and seven months of Android penetration-testing work at bi0s, Amrita Amritapuri. Works across LangGraph, LangChain, Langfuse, RAG, MCP tools, and AI infrastructure, with additional depth in Android and Flutter reverse engineering, binary analysis, Linux security, low-level debugging, and safety-aware automation. Interested in contributing to IBM's open-source document-intelligence ecosystem, including Docling, by building reliable document-to-RAG and enterprise AI workflows.

EDUCATION

B.Tech in Computer Science and Engineering (Artificial Intelligence)

2025–2029 (Expected)

Amrita Vishwa Vidyapeetham · Amrita School of Computing, Amritapuri Campus

EXPERIENCE

Android Pentester — bi0s, Amrita Amritapuri

7-month engagement

- Conducted Android application security analysis, APK inspection, and Flutter reverse engineering through hands-on research and technical writeups.
- Used JADX, Frida, Objection, Reflutter, Ghidra, GDB, Itrace, and Python to inspect application logic, authentication behavior, data flow, and binary transformations.
- Practiced CTF reverse engineering, Linux security, x86-64 assembly, stack and memory inspection, breakpoint analysis, and binary patching.

SELECTED PROJECTS

SentinelOps — Safety-Gated DevOps Agent — Python, LangGraph, LangChain, FastMCP

- Built an incident-response workflow that correlates alerts with logs and recent commits, prepares candidate patches, runs sandbox tests, evaluates confidence and blast radius, and escalates sensitive changes instead of modifying production.

MailDM — Privacy-Focused AI Email Briefing — TypeScript, Discord, Gmail OAuth, Scheduled workflows

- Developed a multi-user application that creates timezone-aware AI summaries from read-only Gmail access and delivers them through Discord direct messages without sending, deleting, archiving, or marking mail as read.

Provider-Neutral Agent Launch — Enterprise AI Architecture — Python, FastAPI, Redis, Prometheus, CI/CD

- Designed a provider-independent agent architecture spanning model providers, runtime, storage, authentication, rate limiting, streaming, testing, monitoring, and deployment.

Open-Source AI Contributions — Cognee, Headroom, NitroStack

- Contributed provider-aware NVIDIA NIM embedding support and tests to Cognee; hardened fork-safe CI workflows; fixed uv-installed Headroom binary detection; and added a Zerogravity MCP sample app to NitroStack.

bi0s Security Writeups and DexProbe — Ghidra, GDB, Itrace, JADX, Frida, Python

- Documented reverse-engineering challenges involving XOR, ROT13, obfuscation, memcmp analysis, Android APK inspection, SSH, Nmap, Netcat, OpenSSL, setuid, and Linux command-line security.

TRAINING AND ACHIEVEMENTS

- Google Cohort 3 / Hack2Skill: completed three Google Cloud hands-on labs involving AI agents, RAG, MCP, BigQuery, and human-in-the-loop workflows; scored 10/10 on all three associated quizzes.
- Summer Analytics 2026: Certificate of Excellence (Top 10%), Certificate of Merit (Top 25%), and capstone completion recognition; focus areas included data science, machine learning, AI, hackathons, Celonis, and process mining.

TECHNICAL SKILLS

Cybersecurity: Android penetration testing, APK analysis, Flutter reverse engineering, dynamic analysis, binary analysis, binary patching, CTFs, Linux security, GDB, x86-64 assembly

Security Tools: JADX, Frida, Objection, Reflutter, Ghidra, GDB, Itrace, Nmap, Netcat, OpenSSL, pwntools, Kali Linux, Ubuntu

AI and Agent Systems: LangGraph, LangChain, Langfuse, Google ADK, RAG, MCP tools, agent orchestration, evaluation and tracing, human-in-the-loop workflows, NLP, machine learning, data science

IBM-Adjacent Focus: Document intelligence, structured document extraction, multimodal AI, enterprise RAG, open-source AI, reliable and responsible AI systems

Programming and Infrastructure: Python, C, C++, Java, JavaScript, FastAPI, Streamlit, Redis, REST APIs, Git/GitHub, CI/CD, Prometheus, Linux

Portfolio: github.com/Ru0k3/portfolio | Public work and writeups available on GitHub